



Weed database development: An updated survey of public weed datasets and cross-season weed detection adaptation

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ABSTRACT

Weeds are a major threat to crop production. Automated innovations for reducing herbicides and labor needed for weeding have become a high priority for sustainable weed management. The current state-of-the-art weeding systems still cannot reliably recognize weeds in changing field conditions for precision weed control. Enhancing weed recognition accentuates the critical need to develop dedicated, labeled weed databases and whereby train advanced AI (artificial intelligence) models while ensuring the robustness of models across diverse field conditions. This study presents an up-to-date survey on publicly available image datasets for weed recognition. Among 36 datasets identified, limitations exist in terms of data variations and distribution shifts, and few of the datasets are suitable for examining the robustness of weed recognition models. A new two-season, eight-class weed dataset is described in this study, comprising two sub-datasets of images collected in the seasons of 2021 and 2022, respectively. Three state-of-the-art deep learning object detectors, i.e., YOLOX, YOLOv8, and DINO, were benchmarked and evaluated for their in-season and cross-season weed detection performance on the dataset. All three models attained in-season detection accuracies of 92% and higher in terms of mAP@50. However substantial accuracy drops of up to 14.5% were observed between in-season and cross-season testing, especially for YOLOX and YOLOv8. Unsupervised domain adaptation based on an implicit instance-invariant network (I³Net) was utilized for improved generalization of the YOLO models. The I³Net-based models resulted in accuracy improvements of 1.4% and 3.3% for YOLOX and YOLOv8, respectively, compared to modeling without domain adaptation, in the cross-season testing. Both the two-season detection dataset¹ and software programs² for weed detection modeling in this study are made publicly available.

1. Introduction

Weeds have a detrimental impact on crop growth and yield, presenting a significant threat to global food security. Among various biotic stressors, weeds are responsible for an average of 34% of crop yield losses worldwide (Oerke, 2006). Left uncontrolled, they have the potential to lead to complete yield loss (Chauhan, 2020). In the United States, the annual economic loss in crop production due to weeds is estimated to be greater than \$30 billion (Pimentel et al., 2000). Since the introduction of trans-genetic, herbicide-resistance crops in the 1990s, weed management has heavily relied on applying herbicides for many field crops (Duke, 2015). Over-reliance on herbicides, however, facilitates the selection of adaptive weed populations and thus the evolution of resistance to herbicides (Délye et al., 2013), substantially increasing

management costs. At the time of writing, 530 unique cases (species × site of action) of herbicide-resistant weeds of 272 species have been reported globally (Heap, 2024). With few herbicides pending release and no new herbicide sites of action introduced for over 30 years (Heap and Duke, 2018), there is an overwhelming need to develop herbicide-reduced or non-chemical weed control tools for sustainable crop production. A recent survey of weed scientists in North America revealed the high research priority of precision weed management and robotics (Brainard et al., 2023).

In recent years, machine vision technologies have grown rapidly and achieved remarkable progress, offering promising solutions for precise weed control. Different vision-guided weed control techniques have been developed (Allmendinger et al., 2022; Machleb et al., 2020; Merfield, 2023), and some of them have been commercially available, e.g.,

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¹ <https://doi.org/10.5281/zenodo.10762138>

² <https://github.com/vicdxxx/CrossSeasonWeedDetection>

laser weeding from Carbon Robotics (Seattle, WA) and See & Spray™ from John Deere (Moline, IL). As artificial intelligence (AI) impacts agriculture, deep learning (DL) techniques, especially convolutional neural networks (CNNs), are ushering in a new paradigm in model development for weed detection and discrimination from crops (Coleman et al., 2022; Lati et al., 2021). Significant research has been conducted on the application of various CNN architectures for weed recognition in the past few years for different cropping systems (Chen et al., 2022; Dang et al., 2023; Gao et al., 2020; Li et al., 2022; Yu et al., 2019), as recently reviewed in Coleman et al. (2022) and Rai et al. (2023a). Despite remarkable progress, challenges persist in reliably and accurately recognizing weeds in changing field conditions by state-of-the-art automated weeding systems with AI models deployed. The erratic performance of commercial weeding systems could occur in fields with dense weed populations (Raja et al., 2019), or when systems were deployed in unseen field conditions according to our personal observations of some latest robotic weeders, which may lead to significant crop damage in intra-row weeding (Lati et al., 2016). The lack of robustness in weed recognition remains a key barrier to practical deployment and wide-scale adoption of automated weeding systems.

Weed recognition challenges are fundamentally due to the significant biological variabilities of weed plants and the presence of many weed species important to crop production. The morphology of weeds varies temporally across different growth stages and seasons and spatially in different field locations. These variations are convoluted with changing field conditions, including lighting, soil types and fertility, and many other uncontrolled factors, adding difficulty to robust weed recognition. To tackle these challenges requires, in addition to pursuing model architecture innovations, creating large-scale, labeled image weed datasets consisting of images acquired in diverse yet relevant field conditions, especially considering recent successes of big data-powered AI models, e.g., ChatGPT (chat generative pre-trained transformer) and SAM (segment anything model), in machine learning and computer vision communities. In surveying public image datasets in precision agriculture, Lu and Sierra (2020) identified 15 weed datasets released from 2015 to 2020. Most of these datasets are comparably small in terms of the number of weed images and species. Coleman et al. (2022) presented an updated list of weed datasets and identified 15 dataset sources, most of which were released before 2022. Despite the updated survey, some weed datasets were not included, and the image acquisition and curation processes important to machine vision applications were not described in adequate detail.

To address the shortage of labeled data, semi-supervised learning methods have been investigated to train models using a small amount of labeled data while leveraging more unlabeled samples (Yang et al., 2022). Khan et al. (2021) built a semi-supervised generative adversarial network (SGAN) for crop and weed classification, acquiring 90% accuracy with over 80% of training images unlabeled, where synthetic images were generated by a modified SGAN to optimize the training process. More studies on GAN-based methods for enhanced weed recognition are reviewed by Lu et al. (2022). Derived from DeepLabV3, SemiWeedNet (Nong et al., 2022) used both labeled and unlabeled images to train a weed segmentation model, and online hard example mining was deployed to concentrate the model on hard-classified pixels, achieving competitive accuracy comparable to that based on fully labeled data. In recognizing weeds in wheat, Liu et al. (2023) proposed a semi-supervised learning method based on consistency regularization and pseudo-labeling while incorporating an attention mechanism, achieving 87.7% testing accuracy when 95% of training samples were unlabeled, compared with the accuracy of 92.2% by supervised learning. Generally, despite data labeling efficiency, semi-supervised learning methods produce lower accuracies than their supervised counterparts.

Data distribution shifts are commonly encountered in deploying machine learning systems, manifested as degraded model performance (Koh et al., 2021). This is conceivably true for weed recognition models

when applied to different field conditions that induce distribution shifts or domain gaps. Models trained on weed data acquired in controlled environments are not likely to generalize well to datasets collected in natural field conditions, and it can also be challenging for weed recognition models to generalize across different growth stages, seasons, and geographical locations. Many existing weed datasets were captured in a particular imaging session or domain (e.g., a specific location and time for limited weed species), which fail to represent a broad array of distributions in diverse field conditions and are not suitable for evaluating the cross-domain robustness of weed recognition models. This probably explains the scarcity of research on model generalization for weed identification (Rai et al., 2023b). In recent years, there has been growing interest in improving the generalization of AI models to achieve real-world deployment success. Various domain adaptation techniques, which involve leveraging labeled source domain data and unlabeled target domain data for model development, have been proposed for enhanced performance in visual recognition tasks (Liu et al., 2023; Oza et al., 2023).

Domain adaptive object detection aims to develop a robust detection model by using label-rich data of the source domain while utilizing the information from label-agnostic or -scarce target domain. Among different adaptation formulations (Oza et al., 2023), unsupervised domain adaptation (UDA), which assumes no annotations are available for the target domain, is more practically useful yet technically challenging. Various UDA approaches have been reported. Domain randomization (Tobin et al., 2017) diversifies the source-domain contexts to mitigate its domain bias. The adversarial domain discriminator (Chen et al., 2018) aims to differentiate the source and target domain. The image-to-image translation transfers the target-domain images into a new domain with a source-like appearance (Rodriguez and Mikolajczyk, 2019; Zhang et al., 2019). Self-training techniques use the target-domain predictions as pseudo labels to learn the target-domain information (RoyChowdhury et al., 2019). As reviewed in Oza et al. (2023), multiple studies (D'Innocente et al., 2020; Kim et al., 2019; Li et al., 2021; Munir et al., 2021; Rodriguez and Mikolajczyk, 2019; Zhao et al., 2020) utilized the pseudo-label-based self-training in cross-domain object detection, combining labeled source-domain images with unlabeled target-domain images. There are typically three levels of feature alignment (at class, image, and instance levels) for utilizing target-domain information. However, many previous studies considered just one or two of these alignments. I³Net (implicit instance-invariant network) (Chen et al., 2021) offers a unification of all three-level feature alignments and can be potentially useful for adapting object detectors to improve the generalization performance of weed detection.

The objective of this study was three-fold, i.e., to: 1) present an up-to-date survey of publicly available weed datasets, 2) curate a new two-season, eight-class weed dataset for cross-season weed detection, and 3) evaluate and improve the generalization performance of state-of-the-art DL object detectors for cross-season weed detection. Specifically, in weed detection experiments, YOLOX (Ge et al., 2021), YOLOv8 (Jocher et al., 2023), and DINO (Zhang et al., 2022) were assessed on the new two-season dataset, and I³Net was leveraged to adapt the two YOLO detectors for potentially improved cross-season weed detection.

2. Public weed dataset survey

Weed datasets were surveyed based on a systematic search in major databases including Google Scholar, ScienceDirect, Web of Science, IEEE Xplore, and arXiv. The search keywords included “weed dataset”, “weed database”, “agriculture”, “machine vision” and “computer vision”. Particularly, to facilitate searching via Google Scholar, the Python library “scholarly”³ was employed to automatically retrieve publication records. A total of 451 records were collected from the literature search,

³ <https://github.com/scholarly-python-package/scholarly>

including 267 articles from Google Scholar, 54 from ScienceDirect, 12 from Web of Science, 106 articles from IEEE Xplore, and 12 articles from arXiv. The initial records were subjected to filtering based on the exclusion criteria, including duplication removal, elimination of review articles, and eligibility verification. Datasets reported in the literature but with the associated website links invalid or broken were also excluded. Table 1 lists the weed image datasets identified.

A total of 36 public weed image datasets have been identified at the time of writing. Given space constraints, a complete list and detailed descriptions of weed datasets are given at <https://github.com/vicdxxx/Weed-Datasets-Survey-2023>, which will be dynamically updated over time. The majority of the identified datasets were captured using RGB (red-green-blue) or color cameras, while some datasets were captured using other imaging sensors (e.g., depth, multispectral, and hyperspectral) or a multimodal approach. Presented below is a brief overview of the datasets which are loosely categorized into different groups based on weed classes.

2.1. Multi-weed-class datasets

The Sugar Beets dataset (Chebrolu et al., 2017) is among the first large multi-class weed datasets, which consists of 9 weed species captured by a ground robot (with artificial lighting) on a sugar beet field in Germany. Multimodal data are included in the dataset, including RGB, NIR (near-infrared), depth (all three types of images were taken from Kinect-v2) images acquired by cameras, point clouds of the plants by laser scanners, and robot-related data (e.g., wheel odometry). The data collection was conducted for two to three days per week over about two months, covering varied crop growth stages (Chebrolu et al., 2017), leading to collected data of approximately 5 TB. The image data consists of two types of pixel-level annotations, including 283 instance annotations of plant species (i.e., sugar beet and 9 weed species) and 12,340 instance annotations of contextual objects (i.e., crop, weeds, and soil background).

The Leaf Counting dataset (Teimouri et al., 2018) consists of RGB or color images labeled in terms of the leaf number of weeds (reflecting plant growth stages), covering 18 weed classes (i.e., weed species or families) in fields across Denmark during three growing seasons. Weeds with 1 leaf to 9 leaves were captured by several hand-held cameras, providing images focusing on an individual plant in each. A total of 9,372 images were annotated at image level annotations for weed species and leaf number.

The Open Plant Phenotype Database (Madsen et al., 2020) consists of 7,590 color images for 47 plant species (including crops and weeds) common in Denmark. The plants were grown under three different conditions for a period from seedling emergence to the stages of 6–8 leaves. Images were captured and collected over three consecutive years by cameras fixed with a top-down view on a railing system under supplemental lighting. Bounding box annotations were provided for different plant species in the dataset, resulting in 64,292 plant instances.

Beck et al., 2021 released a dataset consisting of 14,000 color images for 14 species including crops and weeds common in North America. Images were taken from varied view angles (i.e., profile shots, top-down shots, and angled perspectives), with each image focusing on a single potted plant with a uniform background. Alongside raw image data is the metadata (e.g., camera, date, and capture time) included in the dataset. Despite the seemingly large scale of the dataset, only potted plants were captured under controlled indoor settings, restricting the applicability of this dataset in real-world field scenes.

The AIWeeds dataset (Du et al., 2022) consists of 10,140 color images of the flax and weeds in flax fields taken in different locations. The images were captured under varied sunlight and weather conditions and

for different plant growth stages, but details on platform configuration and imaging devices were not provided. The dataset⁴ was annotated at the image level, and among 14 non-crop classes, the images with non-target plant species or backgrounds were labeled as *negative* samples.

The Weed25 dataset (Wang et al., 2022) consists of 14,035 color images of 25 weed species in East Asia. Most images were acquired under natural, open field conditions at varied growth stages from two to nine leaves, with a small proportion of the weeds cultivated in the pots under controlled conditions of the greenhouse. Images were taken at a close distance (30–50 cm) from the target with each image for one single weed plant. On average, 561 images were annotated for each weed species, coupled with 19,208 bounding box annotations in total.

The CropAndWeed dataset (Steininger et al., 2023) consists of 8,034 plant images captured in diverse Austria field locations for a total of 74 crop and weed species in early growth stages. Images were captured from a top-down view at a height of about 1 m in different environmental conditions (in terms of lighting, weather conditions, and soil types) over a four-year period. Image annotations were performed at different levels including bounding boxes, semantic masks, and stem positions, resulting in about 112 K instances. An extensive set of image metadata about the environmental conditions and camera parameter settings was provided.

The CottonWeedID15 dataset (Chen et al., 2022) contains 5,187 color images of 15 weed classes, which were captured with hand-held cameras in cotton fields in the southern U.S. during two consecutive seasons. The image annotations were done at the image level with each image containing a single class of weeds. Derived from the CottonWeedID15, the CottonWeedDet3 dataset (Rahman et al., 2022) was created for weed detection, which consists of a subset of 848 images for 3 weed species with 1,532 bounding box annotations. Acquired in cotton fields across Mississippi, the CottonWeedDet12 (Lu et al., 2022) is a larger weed detection dataset comprising 5,648 images of 12 weed classes with 9,370 bounding boxes. Alongside the release of all three weed datasets was a benchmark of a suite of deep learning models for weed recognition (Chen et al., 2022; Dang et al., 2023; Rahman et al., 2022).

In addition to weed images, different crops were also captured in some datasets (Gaidel et al., 2023; Madsen et al., 2020; Mylonas et al., 2022; Steininger et al., 2023; Sudars et al., 2020). Sudars et al. (2020) presented a dataset consisting of 1,118 RGB images for 6 crops and 8 weed species, with 7,853 bounding box annotations (441 annotations for crops and 7,442 for weeds). Images were captured with digital cameras in both controlled environments and field conditions. The Eden Library (Mylonas et al., 2022) consisted of 6,428 images captured by proximal sensing platforms and UAVs (unmanned aerial vehicles) in field conditions, for 15 crops and 9 weed species by a handful of imaging devices. Emphasizing on-field image diversity, the database covered different crop health conditions involving 30 disorders, which can be beneficial for model robustness. Metadata was included in the database regarding plant species, imaging hardware specifications and data formats, and a dedicated web application (<https://edenlibrary.ai/eden-library/>) was created for hosting the data and facilitating data reuse, which is dynamically updated. The structured data curation practices with detailed documentation in this study (Mylonas et al., 2022) would be a good reference for the development of large-scale weed databases.

Depth and hyperspectral imagers were also employed for weed imagery. Xu et al. (2022) presented a dataset for two weed species in wheat fields captured using both RGB and RGB-D (depth) cameras. The subset of RGB-D data consisted of three types of images, including RGB, single-channel depth, and three-channel encoded images (for phase, height above ground, and angle with gravity), with 1,288 images for each, according to 21,227 bounding box annotations. Another subset of data

⁴ <https://drive.google.com/drive/folders/1rmvun2wcleDq5fFYR8nA83j6hIHWG9Jz>

Table 1

Major public weed image datasets for weed/weed-crop recognition. Datasets reported in literature but without providing URL links are not included. Links to the datasets summarized in the table are provided at <https://github.com/vicdxxx/Weed-Datasets-Survey-2023>.

References	Image No.	Class No.	Imaging modality	Imaging platform	Recognition task
Dang et al. (2023)	5648	12	RGB	Hand-held camera	Detection
Gaidel et al. (2023)	385	16 (including crops)	HSI	Ground fixed platform	Segmentation
Güldenring et al. (2023)	5510	2	RGB	Ground vehicle	Detection
Kitzler et al. (2023)	2568	18 (10 weeds)	RGB-D	Ground vehicle	Segmentation
Olaniyi et al. (2023)	>10,000	2 (1 weed)	RGB	Hand-held camera	Detection
Rahman et al. (2022)	848	3	RGB	Hand-held camera	Detection
Rai et al. (2023a)	3975	5	RGB	UAV + Hand-held camera	Detection
Steininger et al. (2023)	8034	74 (58 weeds)	RGB	Hand-held camera	Segmentation
Weyler et al. (2023)	2872	5 (1 weed)	RGB	UAV	Segmentation
Alam et al. (2022)	8000	2 (1 weed)	RGB	Hand-held camera	Detection
Chen et al. (2022)	5187	15	RGB	Hand-held camera	Classification
Du et al. (2022)	9324	15 (14 weeds)	RGB	Ground vehicle	Classification
Krestenitis et al. (2022)	201	3	RGB	UAV	Segmentation
Mylonas et al. (2022)	6428	37 (9 weeds)	Multimodal	UAV + Hand-held camera	Segmentation
Teimouri et al. (2022)	1147	2	RGB	Ground vehicle	Detection
Wang et al. (2022)	>10,000	25	RGB	Hand-held camera	Detection
Xu et al. (2022)	1901	2	RGB-D	Ground vehicle	Detection
Ahmad et al. (2021)	462	4	RGB	Hand-held camera	Detection
Beck et al., 2021	>10,000	14 (including crops)	RGB	Ground fixed platform	Classification
Fawakherji et al. (2021)	500	3 (1 weed)	MSI	Ground vehicle	Segmentation
RV et al. (2021)	150	3 (1 weed)	RGB	Hand-held camera	Segmentation
Salazar-Gomez et al. (2022)	4402	2 (1 weed)	RGB	Hand-held camera	Detection
Bender et al. (2020)	>10,000	4 (1 weed)	Multimodal	Ground vehicle	Segmentation
Madsen et al. (2020)	7950	47 (including crops)	RGB	Ground fixed platform	Detection
Sudars et al. (2020)	1118	14 (8 weeds)	RGB	Hand-held camera	Detection
Olsen et al. (2019)	>10,000	8	RGB	Ground vehicle	Classification
Lottes et al. (2018)	1321	2 (1 weed)	RGB	UAV + Ground vehicle	Segmentation
Sa et al. (2018)	>10,000	3 (1 weed)	MSI	UAV	Segmentation
Teimouri et al. (2018)	9372	18 (weeds)	RGB	Hand-held camera	Classification
Chebrolu et al. (2017)	>10,000	10 (9 weeds)	RGB-NIR	Ground vehicle	Classification
Di Cicco et al. (2017)	8878	3 (1 weed)	RGB	Rendering	Segmentation
dos Santos Ferreira et al. (2017)	>10,000	4 (2 weed)	RGB	UAV	Classification
Giselsson et al. (2017)	407	12 (including crops)	RGB	Ground fixed platform	Classification
Lameski et al. (2017)	39	3 (1 weed)	RGB	Hand-held camera	Classification
Sa et al. (2017)	465	3 (1 weed)	MSI	UAV	Classification
Haug and Ostermann (2015)	60	2 (1 weed)	MSI	Ground vehicle	Segmentation

Note: Class No. refers to the total number of weed species, unless stated otherwise in parentheses where the labeled targets may include other objects such as crops and soil background. RGB-D, MSI, HSI, and UAV denote red-green-blue-depth, multispectral imaging, hyperspectral imaging, and unmanned aerial vehicle, respectively.

consisted of 619 RGB images captured at varied view angles with 11,095 bounding box annotations, coupled with 79 depth images in the vertical view. Kitzler et al. (2023) introduced a dataset captured using two RGB cameras in a stereovision setup mounted at a working height of 90 cm on a trolley. Image collection was started right after plant emergence and until plants reached a height of 30 cm. The resulting dataset consisted of 2568 semantic maps annotated by drawing polygons, with 18 classes, including 7 crops and 10 weed species plus soil background, and it also included the metadata for weather conditions, geolocation, and camera parameters. A total of 4,038 annotated instances for 7 crop species and 7,506 instances for 8 weed species were annotated in the dataset. In Gaidel et al. (2023), a hyperspectral image dataset was acquired for 13 classes of crops and 3 classes of weeds with a line-scanning imager, which contained 385 hyperspectral images with 237 bands in the wavelength range of 420–979 nm, stored as NumPy arrays. There are pixel-wise annotations for each instance of plants.

2.2. Uni-weed-class datasets

The uni-weed-class datasets are those that do not differentiate weed species or collectively label weeds as a single class. These datasets can be useful for weed control tactics that target all weeds in the same way but not for high-precision weed-species-specific treatments. Generally, for mechanical weeding, it is adequate to only differentiate between crops and weeds regardless of plant species, but higher-precision weeding requires knowledge of weed species. Several early public weed datasets fall in this category, such as the CWF dataset (Haug and Ostermann, 2015), Carrot-Weed (Lameski et al., 2017), WeedNet (Sa et al., 2017),

Sugar Beet Synthetic dataset (Di Cicco et al., 2017), and WeedMap (Sa et al., 2018). Recently, the sunflower dataset in Fawakherji et al. (2021) consisted of 500 four-channel RGB-NIR images captured at varied crop growth stages, with three-class pixel-wise annotations for crop, weed, and soil background. The Lincoln beet dataset (Salazar-Gomez et al., 2022) contained 4,402 RGB images of weeds and beets captured from three different sugar beet fields (Lincoln, UK), with 39,246 bounding box annotations including 16,399 and 22,847 annotations for beets and weeds, respectively. The PhenoBench dataset (Weyler et al., 2023) consisted of 2,872 RGB images captured in sugar beet fields during two consecutive growth seasons, in which 2,179 images were pixel-wise annotated at both plant-instance (i.e., crops and weeds) and leaf-instance precision levels, particularly with 4,021 and 6,739 annotated instances for plants and leaves, respectively.

One step further in the fine-grained dataset curation is to categorize weeds into broadleaf and grass or other collective groups that demand different weeding tactics, helping optimize weed control (Jin et al., 2022). For instance, ACCase-inhibiting herbicides provide selective control over grassy weeds, while auxin herbicides are commonly used for controlling broadleaf weeds in grass crops (Grossmann, 2010). Example weed datasets following this approach include the soybean crop-weed images (dos Santos Ferreira et al., 2017), the joint-stem detection dataset (Lottes et al., 2018), the MMDWWF dataset (Xu et al., 2022), and the RoboWeedMap dataset (Teimouri et al., 2022). For instance, the RoboWeedMap consisted of 1,147 RGB images captured from over 200 different fields in Denmark, with 19,221 and 41,810 annotations for monocot and dicot weeds, respectively.

2.3. Other weed datasets

To save data collection and annotation efforts, image synthesis has received increasing attention in recent years. The aforementioned Sugar Beet Synthetic dataset (Di Cicco et al., 2017) synthesized 8878 color images, with pixel-wise annotations of three classes including crop, weed, and soil. Procedural generation of computer graphics was utilized by randomly parameterizing the crops and weeds (e.g., the number of leaves, leaf layers, and relative distribution of the plants) and the surroundings (e.g., soil and light conditions) in the target sugar beet field. A leaf kinematic model was utilized, combined with physically based shading techniques, for generating leaves of target plants, and real-world color textures coupled with auxiliary image maps (e.g., ambient occlusion and height), of weeds and crops were deployed. Some published weed datasets contain a significant portion of unlabeled data. The aforementioned CropAndWeed dataset (Steininger et al., 2023) contained a total of 43,814 plant images, but only 8,034 images were labeled. Olaniyi et al. (2023) collected 36,374 RGB images from a variety of maize farmlands for the maize crops and associated weeds. Despite the dataset scale, only 500 images were annotated with 5,764 bounding box instances.

3. Materials and methods

3.1. Dataset curation

The dataset in this study comprised weed images captured in two consecutive seasons of 2021 and 2022. The images captured in 2021 were sampled from the CottonWeedDet12 dataset (Lu, 2023), while the set of images in 2022 was first introduced in the study. Images in both seasons were collected for naturally germinated weeds with hand-held color cameras under natural field conditions in different field sites on research farms at Mississippi State University (Dang et al., 2023). The acquired images were labeled with bounding boxes by trained personnel using the VGG Image Annotator (v2.10) (Dutta and Zisserman, 2019). As done in Dang et al. (2023), the weed instances were labeled by their common names rather than scientific names, and the annotations, exported in .JSON file format, were subjected to quality check before inclusion in the dataset. As a result, a total of 2,717 images collected in 2022 were labeled with 3,959 bounding boxes for 20 known weed species, as shown in Fig. 1, except that 491 bounding boxes existed for unknown weeds in 314 images.

To conduct cross-season weed detection experiments, the images captured in both 2021 and 2022 were used to build a new two-season dataset. Eight abundant weed classes (i.e., Waterhemp, Carpetweed, Morninglory, Goosegrass, Spotted Spurge, Palmer Amaranth, Purslane, and Ragweed) were identified, each of which contained >100 annotation instances for each year. The Year 2021 data, derived from the CottoWeedDet12 dataset (Lu, 2023), contains 4,734 images with 7,664 annotations for the eight weed classes, as opposed to 1,930 images with 3,184 annotations for the Year 2022 data. As such, a new two-season, eight-class weed dataset was created, consisting of 6,664 images with 10,848 bounding boxes, including 7,664 bounding boxes for Year 2021 and 3,184 bounding boxes for Year 2022. The dataset was made publicly available (Lu, 2024). Fig. 2 shows the distribution of images and bounding box numbers of different weed classes for the dataset.

3.2. Detection models in-season and cross-season weed detection

Three object detectors, including YOLOX (Ge et al., 2021), YOLOv8 (Jocher et al., 2023), and DINO (Zhang et al., 2022), were selected to evaluate the performance of both in-season and cross-season detection. YOLOX is a milestone of the one-stage YOLO (You Only Look Once) series, which is presumably the first YOLO model to leverage an anchor-free mechanism (Tian et al., 2019). Utilizing Darknet-53 (Redmon and Farhadi, 2018) as its backbone, YOLOX extracts three-level features,

which are then fused and fed into their innovative decoupled head. These improvements not only expedite its converging speed in the training process but also elevate the detection performance. As the new frontier for object detection, YOLOv8 (Jocher et al., 2023), released in January of 2023, boasts technical features including anchor-free detection mechanisms, efficient convolutional layers, the utilization of feature pyramid networks for recognizing objects of varying sizes, and advanced data augmentation as well as a user-friendly API. Although not formally presented in a peer-reviewed publication, YOLOv8 has gained popularity rapidly and has become a top choice for real-time object detection. Different from CCN-based object detectors like YOLOs, DINO (Zhang et al., 2022) is a transformer-based detector, which evolves from the earlier DETR (DEtection TRansformer) model (Carion et al., 2020), with improved performance through contrastive de-noising training, mixed query selection for anchor utilization, and a look forward twice scheme for box prediction. DINO and its variants have attained state-of-the-art object detection accuracy on the COCO dataset (Lin et al., 2014).

Specifically, YOLOX-large (with 43.6 M parameters) and YOLOv8-large (with 54.2 M parameters), pretrained on the COCO (Lin et al., 2014), and DINO-SwinL-scale4 (with 217.2 M parameters) pretrained on the Object365 (Shao et al., 2019), were chosen for weed detection experiments in this study, which were adapted from their open-source repositories⁵. Fig. 3 shows the experiment pipeline for in-season and cross-season weed detection. In-season modeling refers to the process where the model is trained and tested on the data from the same growth season, while cross-season modeling is for training and testing on the data collected from the different seasons. In either scenario, the corresponding weed dataset was randomly split into train, validation, and test sets according to a ratio of 75%: 5%: 20%. To balance the cross-season model capability, the training set from the Year 2021 data was subsampled in a stratified manner across weed classes to have the same image number as the Year 2022 data, leaving the validation and test sets unchanged, and all the images of the Year 2022 data were utilized in our experiments.

The three weed detectors were fine-tuned with their default hyperparameter settings, with the major settings summarized in Table 2. These settings vary with models, which may not be optimal for weed detection here. For instance, In YOLOX, the BCE (binary cross entropy) and IoU losses were used for classification and regression tasks, respectively. YOLOv8 implemented the DFL (distribution focal loss) (Li et al., 2020) along with CIoU (Complete IoU) (Zheng et al., 2021) for bounding box regression, and VFL (varifocal loss) (Zhang et al., 2021) for its classification task. For DINO, L1 loss and GIoU (Generalized IoU) (Rezatofighi et al., 2019) were used for box regression, with the focal loss (Lin et al., 2017) for classification. No external image augmentation was performed in this study, given the fact that all three detectors have embedded data augmentations. YOLOX and YOLOv8 implement color jitter and mosaic augmentations, while DINO implements random crop and scale augmentation. The original weed images, which were mostly with high resolutions (over 10 megapixels), were resized to the input of 800 × 800 pixels, which is compatible with model architectures and training efficiency. The resizing was done by rescaling the long side of each image to 800 and adding padding to the short side to maintain the original image aspect ratio. All models were trained for 40 epochs with batch size 8, which could adequately converge based on empirical observations.

The weed detection accuracy was quantified in terms of mean Average Precision (mAP) with an IoU (Intersection over Union) threshold of 0.5, i.e., mAP@50. As in our prior work (Chen et al., 2022; Dang et al., 2023), Monte-Carlo cross-validation (Raschka, 2018), with three replications, was implemented to obtain a robust estimate of

⁵ <https://github.com/Megvii-BaseDetection/YOLOX>; <https://github.com/ultralytics/ultralytics>; <https://github.com/IDEA-Research/DINO>

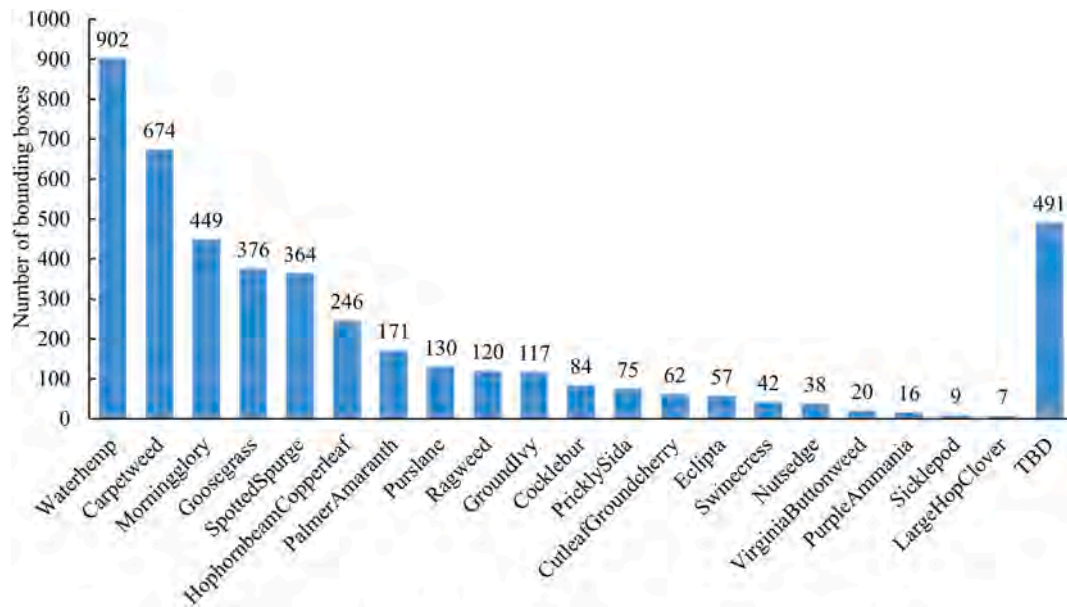


Fig. 1. Number of labeled weed instances for images collected in the season of 2022 for 20 known weed species plus the “TBD” category for unknown weeds that remain to be determined.

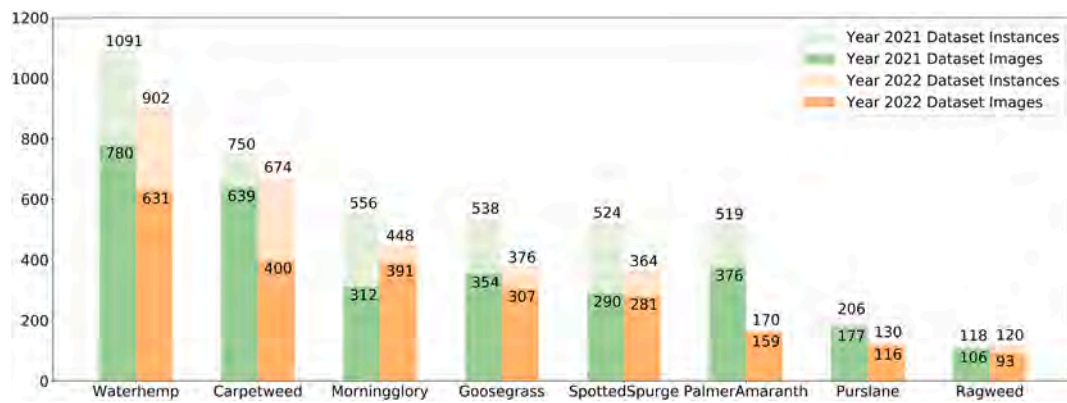


Fig. 2. The image and bounding box number of eight common weed species in the two-season weed dataset.

model performance. In each replication, the corresponding dataset was partitioned into training, validation, and testing sets with a different random seed, and the performance metrics on the test set over three replications were averaged for eventual model assessment. The weed detection modeling was conducted in the framework of PyTorch (version 1.13.1) with Torchvision (version 0.14.1), OpenCV (version 4.7.0), and CUDNN (version 9.2) on an NVIDIA RTX A6000 GPU.

3.3. Unsupervised domain adaptation (UDA)

Given potential dataset shifts across seasons, UDA-based models were built for improved cross-season performance. As a UDA approach, I³Net (Chen et al., 2021) was utilized here to adapt weed detection models for improved generalization. It learns domain-invariant features through three main architectural modules, i.e., category-aware object pattern matching (COPM), dynamic and class-balanced reweighting (DCBR), and regularized joint category alignment (RJCA), as shown in Fig. 4. COPM boosts the cross-domain foreground objects matching guided by categorical information and suppresses uninformative background features. DCBR reweights target samples based on the adaptation difficulty with respect to the intra-domain and intra-class variations. RJCA aligns categories at various high-level layers between source and

target domains with regularizations for feature consistency. More details of I³Net are given in Chen et al. (2021).

The original I³Net (Chen et al., 2021) uses the SSD (single shot multibox detector) (Liu et al., 2016) as the base detection model. In this study, the I³Net was integrated with YOLOX and YOLOv8 for improving their cross-season weed detection performance. For YOLOv8, which consists of 22 nodes (primary modules include multiple layers) in the network architecture⁶ from the initial input node 0 to the final prediction node 21, different nodes were selected for I³Net integration. Specifically, the features from YOLOv8 node 4 and a backbone layer were fed into the pixel-level domain discriminator D_l of COPM, the features from the node 15 were used in DCBR, and the nodes 18 and 21 were fed into the RJCA for cross-domain assignment. Similar integration involving four nodes was done for YOLOX, except that different layers of YOLOX were interfaced with the three modules of I³Net. The output of the YOLOX backbone feature 0 was fed into the domain discriminator D_l of COPM, the head feature 0 into DCBR, and the head features 1 and 2 were connected to RJCA. These connecting nodes in YOLOv8 and YOLOX correspond to the low-level, middle-level, and high-level features of

⁶ <https://github.com/ultralytics/ultralytics/issues/189>

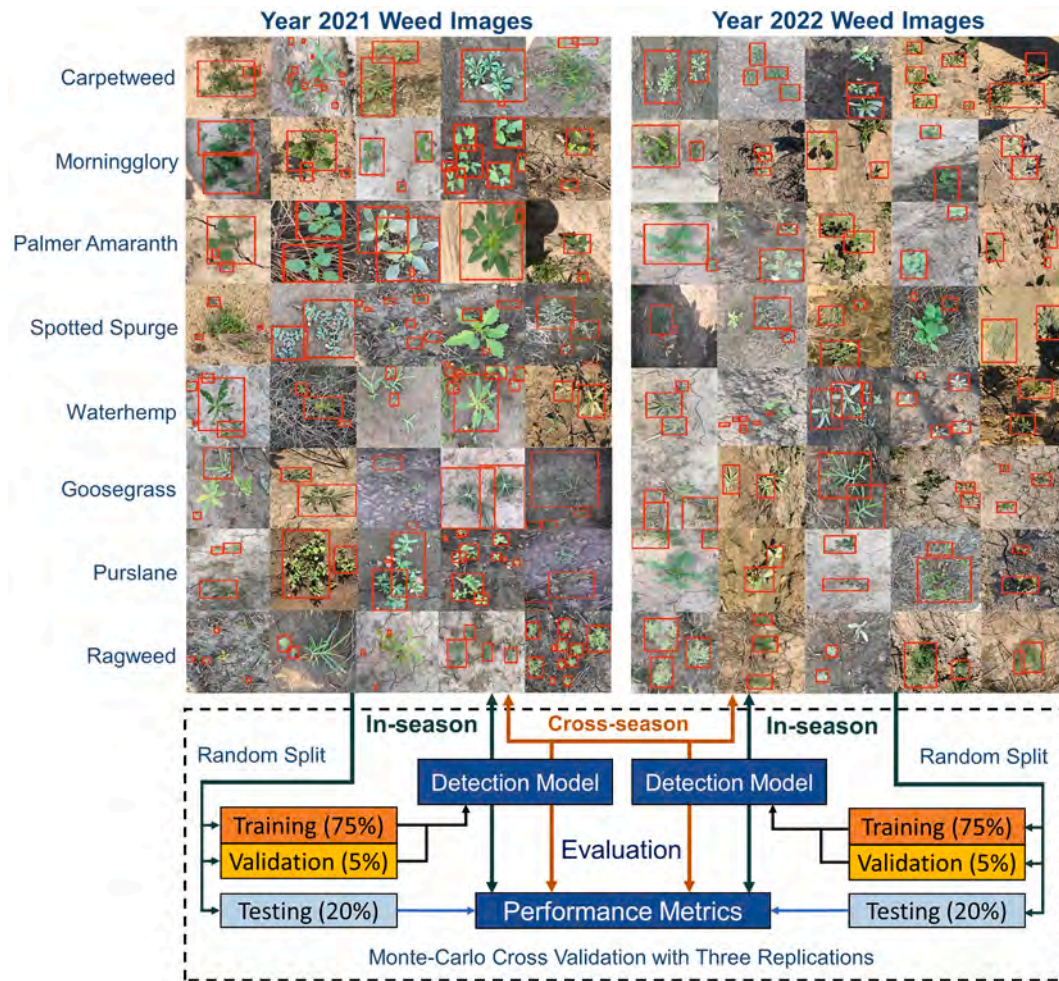


Fig. 3. Flowchart of our cross-season weed detection.

Table 2
Primary hyperparameter settings of the three detection models.

Hyperparameters	Detection models		
	YOLOX	YOLOv8	DINO
Initial learning rate	0.01	0.01	0.0001
Learning rate schedule	Warm up + cosine decay	Warm up + cosine decay	Multistep decay
NMS threshold	0.45	0.7	N/A
Confidence threshold	0.25	0.25	N/A
Weight decay	0.0005	0.0005	0.0001
Image size	800	800	800
Batch size	8	8	8
Optimizer	SGD	SGD	AdamW (Loshchilov and Hutter, 2017)
Classification loss	BCE	VFL (Zhang et al., 2021)	Focal loss
Location loss	IoU	DFL (Li et al., 2020) + CloU (Zheng et al., 2021)	L1 + GloU (Rezatofighi et al., 2019)
Data augmentation	Color jitter + mosaic	Color jitter + mosaic	Random crop and scale

Note: SGD, BCE, VFL, DFL, and NMS denote stochastic gradient descent, binary cross-entropy, varifocal loss, distribution focal loss, and non-maximum suppression, respectively. NMS and confidence thresholds are not applicable for DINO.

different modules of I^3 Net. Unlike training the YOLO models without domain adaptation, a different learning rate scheme was implemented to train the I^3 Net modules. Considering the embedded noise in unpaired target domain images may adversely affect early-stage training, the initial learning rate was set to 0.001 for training the I^3 Net modules, which aimed to ensure the effective learning of source-domain information, being less disturbed by the target domain images, and thereafter the learning rate was reinitialized to 0.01 to enhance feature adaptation to the target domain. Other hyperparameter settings remained as described in Table 2.

4. Results

4.1. In-season and cross-season weed detection

Fig. 5 shows the weed detection accuracies without performing domain adaptation. For the in-season detection models, high accuracies were obtained with mAP@50 ranging from 92.0% to 98.7%, with the best accuracy obtained by DINO, and YOLOX and YOLOv8 achieved comparable accuracies. The three detectors performed consistently better on the Year 2022 data than on the Year 2021 data, which may be explained by the higher diversity observed in the Year 2021 data, requiring higher learning capability of models. The exact reason for differing performance in the two sub-datasets remains to be ascertained by analyzing the characteristics of the two-season dataset to reveal those influencing the detection. In cross-season modeling, substantial accuracy drops were observed for all three models trained on the 2022 data

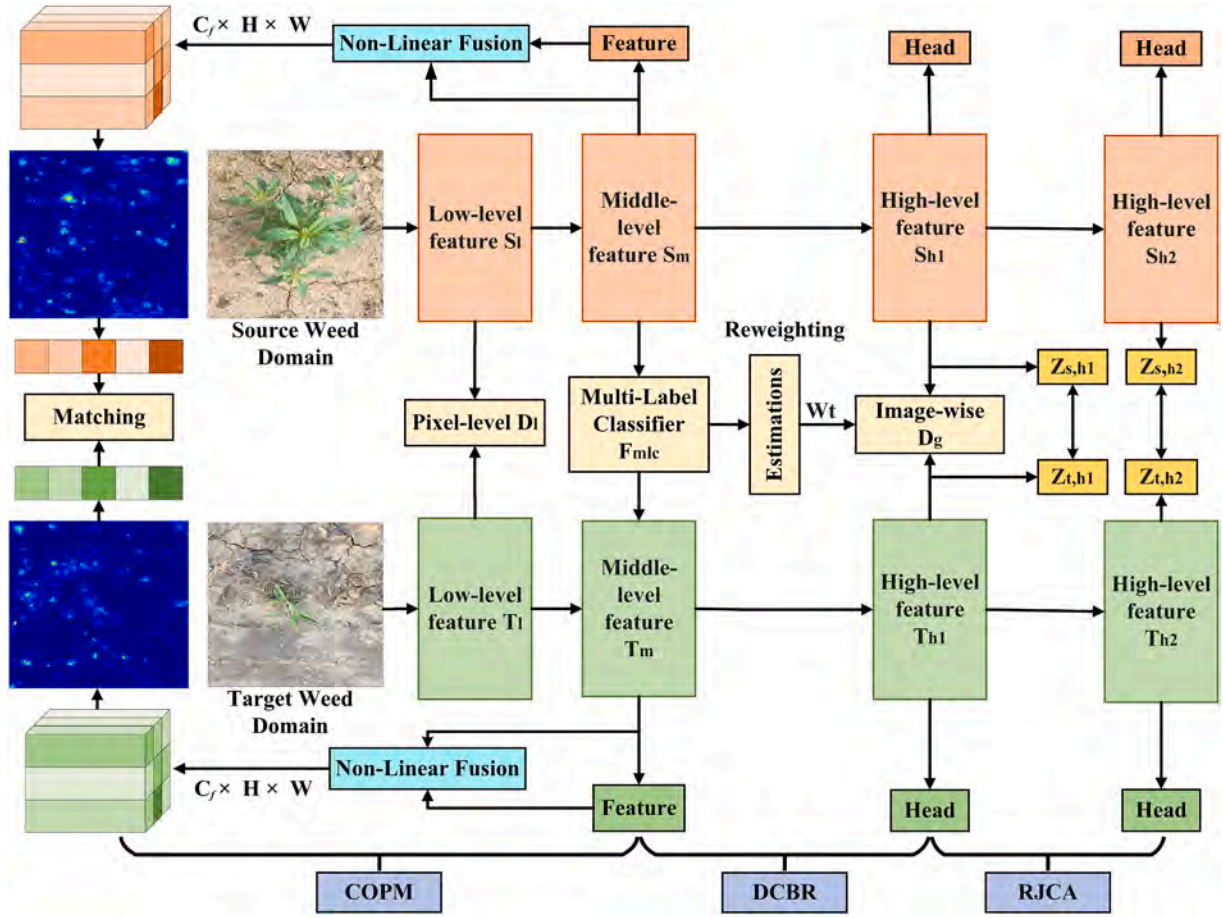


Fig. 4. Domain adaptation framework I^3 Net used in this study. COPM, DCBR, and RJCA denote the modules of category-aware object pattern matching, dynamic and class-balanced reweighting, and regularized joint category alignment, respectively, in Chen et al. (2021).

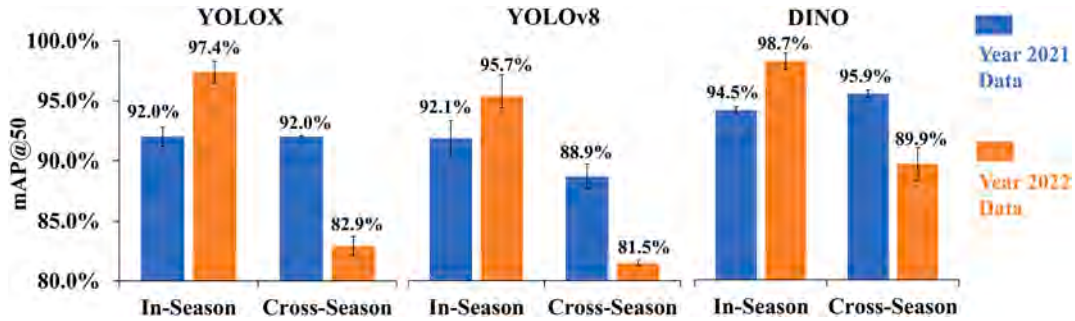


Fig. 5. Weed detection accuracy by three detectors. In-season and cross-season metrics refer to the accuracies of models trained and tested on the data from the same and different seasons, respectively, and the error bar indicates the standard error over three modeling replications. The year legends are for the training data.

and tested on the 2021 data, in comparison with in-season models for the 2022 data, and the drops were 14.5%, 14.2%, and 8.8% for YOLOX, YOLOv8, and DINO, respectively. On the other hand, the models trained on the Year 2021 data generalized better to the Year 2022 data, except that an accuracy drop of 3.2% was observed for YOLOv8 compared to the corresponding in-season modeling. Overall, these results suggest that the YOLOv8 and YOLOX models had weaker cross-season generalization ability than the DINO model. The latter achieved improvements ranging from 3.8% to 8.4% in cross-season weed detection compared to the two YOLO models. The improved performance agrees with the fact that the transformer-based DINO model has about four times the learnable parameters compared to the YOLO models, implying higher model complexity and better learning compacity for the intended task.

Fig. 6 shows differing predictions of in-season and cross-season YOLOv8 models for the same images.

4.2. I^3 Net-based weed detection

Fig. 7 compares the weed detection performance of YOLO models with and without I^3 Net-based domain adaptation. The I^3 Net-based models achieved higher accuracies in all cross-season scenarios, and also higher or comparable accuracies in in-season tests. Particularly for cross-season modeling, the I^3 Net-based YOLOX (trained on the Year 2021 data and tested on the Year 2022 data), improved the detection by 1.1% mAP@50 (from 92.0% to 93.1%) compared to the original YOLOX without I^3 Net, and an improvement of 1.4% mAP@50 (from 82.9% to



Fig. 6. Example predictions by YOLOv8 trained on the Year 2021 data and tested the images from the same season (in-season) and trained on the Year 2022 data and tested on the Year 2022 data (cross-season). The percentage and label attached to each bounding box indicate the detection confidence and the predicted weed class, respectively.

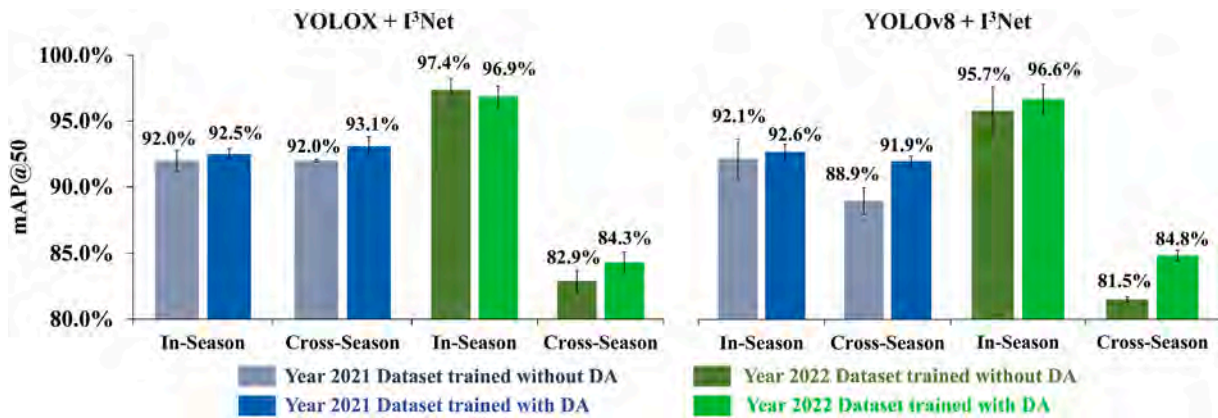


Fig. 7. Weed detection accuracies of YOLOX and YOLOv8 with and without I³Net-based domain adaptation (DA). In-season and cross-season metrics refer to the accuracies of models trained and tested on the data from the same and different seasons, respectively, and the error bar indicates the standard error over three modeling replications.

84.3%) was attained for the model trained on the Year 2022 and tested on the Year 2021 data. The YOLOv8 with I³Net resulted in an improvement of 3% mAP@50 (from 88.9% to 91.9%) in the cross-season detection when trained on the Year 2021 and tested on the Year 2022 data, and on the other hand, it yielded an improvement of 3.3% mAP@50 (from 81.5% to 84.8%) while trained on the Year 2022 and tested on the Year 2021 data. For both I³Net-based YOLOX and YOLOv8, which were trained on the Year 2022 data and tested on the data from the same and different seasons, there were still performance gaps >10% mAP@50. Nonetheless, the cross-season performance improvements ranging from 1.1% to 3.3% mAP@50, relative to the corresponding accuracies obtained without domain adaptation, were found statistically significant (over three replications) according to a paired-sample *t*-test at a significance level $\alpha = 0.05$. These results demonstrated the efficacy of I³Net-based domain adaptation for improving cross-season generalization of YOLO models.

5. Discussion

The current publicly available weed datasets are valuable resources for weed recognition research. However, a few limitations or gaps need to be identified based on the weed dataset survey presented in this study. As pointed out by Lu and Sierra (2020), despite the recent efforts in weed image database development, none of the published weed datasets are comparable to those impactful datasets in computer vision such as ImageNet (Deng et al., 2009), COCO (Lin et al., 2014), and Open Images (Kuznetsova et al., 2020), in terms of image or instance numbers, highlighting the dire need of continued data curation efforts. Imagery data that capture the biological variability of weeds/crops and field condition variations are crucial for developing robust weed recognition models. Although many weed datasets contained images collected in varying conditions, few datasets are dedicated to capturing the plant and field factors (e.g., growth stages and seasons, soil conditions) in a systematic manner, and in this regard their applicability to tackle domain

shift problems in weed recognition is still limited. Following the weed dataset survey, this study presents a new two-season multi-class weed dataset, which has been made publicly available (Lu, 2024), and an evaluation of cross-season weed detection performance.

The new two-season weed dataset in the study represents an initial step for constructing a testbed to clearly reflect the across-season domain shift issues that have been less investigated in the literature on weed detection. Addressing the issues is necessary for developing robust weed recognition models across seasons and, furthermore, for creating practically useful weeding systems. For this purpose, three representative detection models were experimented on the two-season dataset, demonstrating the generalization issues due to the domain shift, especially for the two YOLO models, although the present dataset is still limited in the number of images, bounding boxes, and weed classes, as well as seasons. A more large-scale, diverse multi-season dataset is needed for extensive evaluation of cross-season model robustness. In addition to season factors, practically useful models are also expected to be robust to many other important temporal and spatial factors such as plant growth stages, soil types, field locations, etc., which demand dedicated data curation efforts in future research. Experimented on the two-season dataset, the I³Net-based domain adaptation approach positively impacted the cross-season generalization of YOLO models, but there is room for further reducing performance drops and achieving higher accuracies comparable to that of in-season modeling in some scenarios (Fig. 7). Although coarse-to-fine feature adaption (Zheng et al., 2020) is a widely acknowledged paradigm for cross-domain object detection, there has been a lack of dedicated research to introduce hierarchy feature adaptation techniques into the context of cross-season weed detection, leaving a largely unexplored area for improvements. The I³Net-based domain adaptation technique experimented in this study represents a promising starting point for exploring coarse-to-fine feature adaptation approaches, opening new opportunities in the field.

In addition to cross-season performance drops in weed detection in this study, performance drops of up to 45% accuracy were observed in a recent study on cross-domain crop-weed segmentation (Weyler et al., 2023). The performance gaps were reduced to around 10% by introducing unlabeled plant images from additional agricultural fields (beyond the source and target domains) to diversify source-domain images and enhance the training process associated with the approximate annotations generated by heuristic methods (e.g., color indices and crop row arrangement). Many potential methods are available to address data shift issues in cross-domain modeling tasks, particularly advanced multimodal generative models, such as stable diffusion (Rombach et al., 2022), and DALL-E 3 (<https://openai.com/dall-e-3>), among other generative modeling techniques (Lu et al., 2022). These models can generate realistic high-resolution/-fidelity images that can be used to diversify the source datasets and enhance target-domain data. Furthermore, specific light conditions and accurately assigned object locations (Zhang et al., 2023) can be achieved to provide precise control over image synthesis, offering potential means for boosting cross-domain detection.

Future research into the quantitative assessment of domain gaps or shifts between training and testing data can be beneficial for developing effective methods for improved cross-domain performance and deciding on whether domain adaptation/generalization is needed on unseen data (Doan et al., 2023). The current evaluation metrics (Kynkäänniemi et al., 2019) focus on image fidelity and reality, which contribute to the assessment of image generation by rapidly growing diffusion models instead of evaluating the domain gaps. On the other hand, the human evaluation protocol, which is widely adopted to distinguish synthetic images, like side-by-side image comparison (Wang et al., 2023), can be a useful strategy to measure the domain gaps before more solid domain-gap metrics are discovered.

6. Conclusion

This study surveyed publicly available weed datasets directed at machine vision-based weed management, identifying 36 datasets captured for different weeds and crops in varied imaging conditions. These datasets are valuable resources for weed recognition research. However, few datasets are large-scale in comparison to those open datasets widely used in the computer vision community because of unique challenges in weed dataset curation. More dedicated efforts are needed to create large-scale weed datasets while capturing important domain variations, spatially and temporally, so as to enable the development of robust weed recognition models. In this regard, a new two-season, eight-class dataset captured in natural field conditions was presented in this study, consisting of 4,734 images with 7,664 bounding box annotations and 1,930 images with 3,184 bounding boxes, captured in two consecutive seasons of season of 2021 and 2022, respectively. Three object detectors (YOLOX-large, YOLOv8-large, and DINO-SwinL-scale4) were evaluated on the dataset for both in-season and cross-season weed detection. The cross-season accuracy in terms of mAP@50 was found to drop substantially by up to 14.5% compared to that in the corresponding in-season detection, implying cross-season data domain shifts. The DINO compared favorably to the two YOLO detectors in cross-season generalization performance. A UDA approach based on I³Net was employed to adapt the two YOLO detectors to reduce performance drops across seasons. The I³Net-based YOLOX and YOLOv8 models attained improvements of up to 1.4% and 3.3% mAP@50 compared to their counterparts without I³Net, which were statistically significant. Extensive research has yet to be done to develop more effective approaches to mitigate the cross-season performance drops. In addition to pursuing algorithmic innovations in domain adaptation/generalization, having larger sets of representative imagery will be beneficial and examined in our future work.

CRedit authorship contribution statement

Boyang Deng: Writing – original draft, Formal analysis, Data curation. **Yuzhen Lu:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Jiajun Xu:** Writing – review & editing.

Data availability

The weed dataset and software programs in this study are publicly available at <https://doi.org/10.5281/zenodo.10762138> and <https://github.com/vicdxxx/CrossSeasonWeedDetection>, respectively.

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References

- Ahmad, A., Saraswat, D., Aggarwal, V., Etienne, A., Hancock, B., 2021. Performance of deep learning models for classifying and detecting common weeds in corn and soybean production systems. *Comput. Electron. Agric.* 184, 106081.
- Alam, M.S., Alam, M., Tufail, M., Khan, M.U., Güneş, A., Salah, B., Nasir, F.E., Saleem, W., Khan, M.T., 2022. TobSet: A new tobacco crop and weeds image dataset and its utilization for vision-based spraying by agricultural robots. *Appl. Sci.* 12 (3), 1308.
- Allmendinger, A., Spaeth, M., Saile, M., Peteinatos, G.G., Gerhards, R., 2022. Precision chemical weed management strategies: A review and a design of a new CNN-based modular spot sprayer. *Agronomy* 12 (7), 1620.
- Beck, Michael A., Liu, Chen-Yi, Bidinosti, Christopher P., Henry, Christopher J., Godee, Cara M., Ajmani, Manisha, 2021. Presenting an extensive lab-and field-image dataset of crops and weeds for computer vision tasks in agriculture. *arXiv preprint arXiv:2108.05789*.

- Bender, Asher, Whelan, Brett, Sukkari, Salah, 2020. A high-resolution, multimodal data set for agricultural robotics: a Ladybird's-eye view of Brassica. *J. Field Robot.* 37 (1), 73–96.
- Brainard, D.C., Haramoto, E.R., Leon, R.G., Kells, J.J., Wychen, L.R.V., Devkota, P., Jugulum, M., Barney, J.N., 2023. A survey of weed research priorities: key findings and future directions. *Weed Sci.* 71 (4), 330–343.
- Carion, Nicolas, Massa, Francisco, Synnaeve, Gabriel, Usunier, Nicolas, Kirillov, Alexander, Zagoruyko, Sergey, 2020. End-to-end object detection with transformers. In: Paper Presented at the European Conference on Computer Vision.
- Chauhan, B.S., 2020. Grand challenges in weed management. *Front. Agronomy* 1, 3.
- Chebrolov, Nived, Lottes, Philipp, Schaefer, Alexander, Winterhalter, Wera, Burgard, Wolfram, Stachniss, Cyrill, 2017. Agricultural robot dataset for plant classification, localization and mapping on sugar beet fields. *Int. J. Robot. Res.* 36 (10), 1045–1052.
- Chen, Y., Li, W., Sakaridis, C., Dai, D., Van Gool, L., 2018. Domain adaptive faster r-cnn for object detection in the wild. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 3339–3348.
- Chen, Chaoqi, Zheng, Zebiao, Huang, Yue, Ding, Xinghao, Yu, Yizhou, 2021. I3net: Implicit instance-invariant network for adapting one-stage object detectors. In: Paper Presented at the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition.
- Chen, Dong, Lu, Yuzhen, Li, Zhaojian, Young, Sierra, 2022. Performance evaluation of deep transfer learning on multi-class identification of common weed species in cotton production systems. *Comput. Electron. Agric.* 198, 107091.
- Coleman, G.R.Y., Hu, K., Sharpe, S.M., Schumann, A.W., Wang, Z., Bagavathiannan, M. V., Boyd, N.S., Walsh, M.J., 2022. Weed detection to weed recognition: reviewing 50 years of research to identify constraints and opportunities for large-scale cropping systems. *Weed Tech.* 36 (6), 741–757.
- Dang, Fengying, Chen, Dong, Lu, Yuzhen, Li, Zhaojian, 2023. YOLOWeeds: a novel benchmark of YOLO object detectors for multi-class weed detection in cotton production systems. *Comput. Electron. Agric.* 205, 107655.
- Délye, C., Jasieniuk, M., Le Corre, V., 2013. Deciphering the evolution of herbicide resistance in weeds. *Trends Genet.* 29 (11), 649–658.
- Deng, J., Dong, W., Socher, R., Li, L.J., Li, K., Li, F.F., 2009. In: Imagenet: A large-scale hierarchical image database. *IEEE Conference on Computer Vision and Pattern Recognition*, pp. 248–255.
- Di Cicco, Maurizio, Potena, Ciro, Grisetti, Giorgio, Pretto, Alberto, 2017. Automatic model based dataset generation for fast and accurate crop and weeds detection. In: Paper presented at the 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).
- D'Innocente, Antonio, Borlino, Francesco, Cappio, Bucci, Silvia, Caputo, Barbara, Tommasi, Tatiana, 2020. One-shot unsupervised cross-domain detection. In: Paper presented at the Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XVI, p. 16.
- Doan, Anh-Dzung, Nguyen, Bach Long, Gupta, Surabhi, Reid, Ian, Wagner, Markus, Chin, Tat-Jun, 2023. Assessing domain gap for continual domain adaptation in object detection. *arXiv preprint arXiv:2302.10396*.
- dos Santos Ferreira, A., Freitas, D.M., da Silva, Pistori, H., Folhes, M.T., 2017. Weed detection in soybean crops using ConvNets. *Comput. Electron. Agric.* 143, 314–324.
- Du, Yayun, Zhang, Guofeng, Tsang, Darren, Jawed, Mohammad, Khalid, 2022. Deep-cnn based robotic multi-class under-canopy weed control in precision farming. In: Paper Presented at the 2022 International Conference on Robotics and Automation (ICRA).
- Duke, S.O., 2015. Perspectives on transgenic, herbicide-resistant crops in the United States almost 20 years after introduction. *Pest Manag. Sci.* 71 (5), 652–657.
- Dutta, Abhishek, Zisserman, Andrew, 2019. The VIA annotation software for images, audio and video. In: Paper Presented at the Proceedings of the 27th ACM International Conference on Multimedia.
- Fawakherji, Mulham, Potena, Ciro, Pretto, Alberto, Bloisi, Domenico D., Nardi, Daniele, 2021. Multi-spectral image synthesis for crop/weed segmentation in precision farming. *Robot. Auton. Syst.* 146, 103861.
- Gaidel, A.V., Podlipnov, V.V., Ivliev, N.A., Paringer, R.A., Ishkin, P.A., Mashkov, S.V., Skidanov, R.V., 2023. Agricultural plant hyperspectral imaging dataset. *Компьютерная оптика* 47 (3), 442–450.
- Gao, Junfeng, French, Andrew P., Pound, Michael P., He, Yong, Pridmore, Tony P., Pieters, Jan G., 2020. Deep convolutional neural networks for image-based *Convolvulus sepium* detection in sugar beet fields. *Plant Methods* 16 (1), 1–12.
- Ge, Zheng, Liu, Songtao, Wang, Feng, Li, Zeming, Sun, Jian, 2021. YOLOX: Exceeding yolo series in 2021. *arXiv preprint arXiv:2107.08430*. <https://arxiv.org/pdf/2107.08430.pdf>.
- Giselsson, Thomas, Mosgaard, Jørgensen, Rasmus, Nyholm, Jensen, Peter, Kryger, Dyrmann, Mads, Midtby, Henrik Skov, 2017. A public image database for benchmark of plant seedling classification algorithms. *arXiv preprint arXiv:1711.05458*.
- Grossmann, K., 2010. Auxin herbicides: current status of mechanism and mode of action. *Pest Manag. Sci.* 66 (2), 113–120.
- Güldenring, R., Van Evert, Nalpanidis, L., 2023. RumexWeeds: A grassland dataset for agricultural robotics. *J. Field Robot.* 40 (6), 1639–1656.
- Haug, Sebastian, Ostermann, Jörn, 2015. A crop/weed field image dataset for the evaluation of computer vision based precision agriculture tasks. In: Paper presented at the Computer Vision–ECCV 2014 Workshops: Zurich, Switzerland, September 6–7 and 12, 2014, Proceedings, Part IV, p. 13.
- Heap, I., Duke, S.O., 2018. Overview of glyphosate-resistant weeds worldwide. *Pest Manag. Sci.* 74 (5), 1040–1049.
- Jin, X., Bagavathiannan, M., Mity, A., Chen, Y., Yu, J., 2022. Deep learning for detecting herbicide weed control spectrum in turfgrass. *Plant Methods* 18 (1), 94.
- Jocher, G., Chaurasia, A., Qiu, J., 2023. YOLO by Ultralytics (Version 8.0.0) [Computer Software]. <https://github.com/ultralytics/ultralytics>.
- Khan, S., Tufail, M., Khan, M.T., Khan, Z.A., Iqbal, J., Alam, M., 2021. A novel semi-supervised framework for UAV based crop/weed classification. *PLoS One* 16 (5), e0251008.
- Kim, Seunghyeon, Choi, Jaehoon, Kim, Taekyung, Kim, Changick, 2019. Self-training and adversarial background regularization for unsupervised domain adaptive one-stage object detection. In: Paper Presented at the Proceedings of the IEEE/CVF International Conference on Computer Vision.
- Kitzler, Florian, Barta, Norbert, Neugschwandtner, Reinhard W., Gronauer, Andreas, Motsch, Viktoria, 2023. WE3DS: an RGB-D image dataset for semantic segmentation in agriculture. *Sensors* 23 (5), 2713.
- Koh, Pang Wei, Sagawa, Shiori, Marklund, Henrik, Xie, Sang Michael, Zhang, Marvin, Balsubramani, Akshay, Hu, Weihua, Yasunaga, Michihiro, Phillips, Richard, Lanus, Gao, Irena, 2021. Wilds: A benchmark of in-the-wild distribution shifts. In: Paper Presented at the International Conference on Machine Learning.
- Krestenitis, M., Raptis, E.K., Kapoutsis, A.C., Ioannidis, K., Kosmatopoulos, E.B., Vrochidis, S., Kompatsiaris, I., 2022. CoFly-WeedDB: A UAV image dataset for weed detection and species identification. *Data in Brief* 45, 108575.
- Kuznetsova, A., Rom, H., Alldrin, N., Uijlings, J., Krasin, I., Pont-Tuset, J., Kamali, S., Popov, S., Mallocci, M., Duerig, T., Ferrari, V., 2020. The open images dataset v4: Unified image classification, object detection, and visual relationship detection at scale. *Int. J. Comput. Vis.* 128 (7), 1956–1981.
- Kynkäänniemi, Tuomas, Karras, Tero, Laine, Samuli, Lehtinen, Jaakko, Aila, Timo, 2019. Improved precision and recall metric for assessing generative models. *Adv. Neural Inf. Process. Syst.* 32.
- Lameski, Petre, Zdravetski, Eftim, Trajkovik, Vladimir, Kulakov, Andrea, 2017. Weed detection dataset with RGB images taken under variable light conditions. In: Paper Presented at the ICT Innovations 2017: Data-Driven Innovation. 9th International Conference, ICT Innovations 2017, Skopje, Macedonia, September 18–23, 2017, Proceedings, p. 9.
- Li, Xiang, Wang, Wenhui, Wu, Lijun, Chen, Shuo, Hu, Xiaolin, Li, Jun, Tang, Jinhui, Yang, Jian, 2020. Generalized focal loss: learning qualified and distributed bounding boxes for dense object detection. *Adv. Neural Inf. Process. Syst.* 33, 21002–21012.
- Lati, R.N., Rasmussen, J., Andujar, D., Dorado, J., Berge, T.W., Wellhausen, C., Pflanz, M., Nordmeyer, H., Schirrmann, M., Eizenberg, H., 2021. Site-specific weed management—constraints and opportunities for the weed research community: Insights from a workshop. *Weed Res.* 61 (3), 147–153.
- Lati, R.N., Siemens, M.C., Rachuy, J.S., Fennimore, S.A., 2016. Intra-row weed removal in broccoli and transplanted lettuce with an intelligent cultivator. *Weed Technology* 30 (3), 655–663.
- Li, Xianfeng, Chen, Weijie, Xie, Di, Yang, Shicai, Yuan, Peng, Pu, Shiliang, Zhuang, Yueting, 2021. A free lunch for unsupervised domain adaptive object detection without source data. In: Paper Presented at the Proceedings of the AAAI Conference on Artificial Intelligence.
- Li, Yong, Guo, Zhiqiang, Shuang, Feng, Zhang, Man, Li, Xiuhua, 2022. Key technologies of machine vision for weeding robots: a review and benchmark. *Comput. Electron. Agric.* 196, 106880.
- Lin, Tsung-Yi, Maire, Michael, Belongie, Serge, Hays, James, Perona, Pietro, Ramanan, Deva, Dollár, Piotr, Zitnick, C. Lawrence, 2014. Microsoft coco: Common objects in context. In: Paper Presented at the Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6–12, 2014, Proceedings, Part V, p. 13.
- Lin, T.-Y., Goyal, P., Girshick, R., He, K., Dollár, P., 2017. Focal loss for dense object detection. In: Paper Presented at the Proceedings of the IEEE International Conference on Computer Vision.
- Liu, Wei, Angelov, Dragomir, Erhan, Dumitru, Szegedy, Christian, Reed, Scott, Fu, Cheng-Yang, Berg, Alexander C., 2016. Ssd: Single shot multibox detector. In: Paper Presented at the Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part I, p. 14.
- Liu, T., Jin, X., Zhang, L., Wang, J., Chen, Y., Hu, C., Yu, J., 2023. Semi-supervised learning and attention mechanism for weed detection in wheat. *Crop Prot.* 174, 106389.
- Loshchilov, Ilya, Hutter, Frank, 2017. Decoupled weight decay regularization. *arXiv*. <https://arxiv.org/pdf/1711.05101.pdf>. preprint arXiv:1711.05101.
- Lottes, Philipp, Behley, Jens, Chebrolov, Nived, Milioto, Andres, Stachniss, Cyrill, 2018. Joint stem detection and crop-weed classification for plant-specific treatment in precision farming. In: Paper Presented at the 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).
- Lu, Y., Sierra, Y., 2020. A survey of public datasets for computer vision tasks in precision agriculture. *Comput. Electron. Agric.* 178, 105760.
- Lu, Y., 2024. Two-Season-WeedDet8: a two-season, 8-class dataset for cross-season weed detection generalization evaluation [Data set]. *Zenodo*. <https://doi.org/10.5281/zenodo.10762138>.
- Lu, Y., Chen, D., Olaniyi, E., Huang, Y., 2022. Generative adversarial networks (GANs) for image augmentation in agriculture: a systematic review. *Comput. Electron. Agric.* 200, 107208.
- Machleb, J., Peteinatos, G.G., Kollenda, B.L., Andújar, D., Gerhards, R., 2020. Sensor-based mechanical weed control: Present state and prospects. *Computers and electronics in agriculture* 176, 105638.
- Madsen, Leminen, Simon, Mathiasen, Kopp, Solvejg, Dyrmann, Mads, Laursen, Morten, Stigaard, Paz, Laura-Carlota, Jørgensen, Rasmus, Nyholm, 2020. Open plant phenotype database of common weeds in Denmark. *Remote Sens.* 12 (8), 1246.
- Merfield, C.N., 2023. Integrated weed management in organic farming. In: Sarathchandran, Unni, M.R., Thomas, Sabu, Meena, D.K. (Eds.), *Organic Farming*, Second Edition. Woodhead Publishing, pp. 31–109.

- Munir, Muhammad Akhtar, Khan, Muhammad Haris, Sarfraz, M., Ali, Mohsen, 2021. Ssal: synergizing between self-training and adversarial learning for domain adaptive object detection. *Adv. Neural Inf. Process. Syst.* 34, 22770–22782.
- Mylonas, Nikos, Malounas, Ioannis, Mouseti, Sofia, Vali, Eleanna, Espejo-Garcia, Borja, Fountas, Spyros, 2022. Eden library: a long-term database for storing agricultural multi-sensor datasets from UAV and proximal platforms. *Smart Agric. Technol.* 2, 100028.
- Lu, Y., 2023. CottonWeedDet12: a 12-class weed dataset of cotton production systems for benchmarking AI models for weed detection [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.7535814>.
- Nong, C., Fang, X., Wang, J., 2022. Semi-supervised learning for weed and crop segmentation using UAV imagery. *Front. Plant Sci.* 13, Article 927368.
- Oerke, E.-C., 2006. Crop losses to pests. *J. Agric. Sci.* 144 (1), 31–43.
- Olaniyi, Olayemi Mikail, Salaudeen, Muhammadu Tajudeen, Daniya, Emmanuel, Abdullahi, Ibrahim Mohammed, Folorunso, Taliahi Abiodun, Bala, Jibril Abdullahi, Nuhu, Bello Kontagora, Adedigba, Adeyinka Peace, Oluwale, Blessing Israel, Bankole, Abdullah Oreoluwa, 2023. Development of maize plant dataset for intelligent recognition and weed control. *Data Brief* 47, 109030.
- Olsen, Alex, Konovalov, Dmitry A., Philippa, Bronson, Ridd, Peter, Wood, Jake C., Johns, Jamie, Banks, Wesley, Girgenti, Benjamin, Kenny, Owen, Whinney, James, 2019. DeepWeeds: a multiclass weed species image dataset for deep learning. *Sci. Rep.* 9 (1), 2058.
- Oza, Poojan, Sindagi, Vishwanath A., Sharmini, Vibashan Vishnukumar, Patel, Vishal M., 2023. Unsupervised domain adaptation of object detectors: A survey. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence*.
- Pimentel, D., Lach, L., Zuniga, R., Morrison, D., 2000. Environmental and economic costs of nonindigenous species in the United States. *BioScience* 50 (1), 53–65.
- Rahman, A., Lu, Y., Wang, H., 2022. Deep neural networks for weed detections towards precision weeding. In: *2022 ASABE Annual International Meeting*. American Society of Agricultural and Biological Engineers, p. 1.
- Rai, Nitin, Mahecha, Maria Villamil, Christensen, Annika, Quanbeck, Jamison, Zhang, Yu, Howatt, Kirk, Ostlie, Michael, Sun, Xin, 2023a. Multi-format open-source weed image dataset for real-time weed identification in precision agriculture. *Data Brief* 51, 109691.
- Rai, Nitin, Zhang, Yu, Ram, Billy G., Schumacher, Leon, Yellavajjala, Ravi K., Bajwa, Sreekala, Sun, Xin, 2023b. Applications of deep learning in precision weed management: a review. *Comput. Electron. Agric.* 206, 107698.
- Raja, R., Slaughter, D.C., Fennimore, S.A., Nguyen, T.T., Vuong, V.L., Sinha, N., Tourte, L., Smith, R.F., Siemens, M.C., 2019. Crop signalling: A novel crop recognition technique for robotic weed control. *Biosyst. Eng.* 187, 278–291.
- Raschka, Sebastian, 2018. Model evaluation, model selection, and algorithm selection in machine learning. *arXiv preprint arXiv:1811.12808*.
- Redmon, J., Farhadi, A., 2018. Yolov3: An incremental improvement. *arXiv preprint arXiv:1804.02767*.
- Rezatoghli, Hamid, Tsoi, Nathan, Gwak, JunYoung, Sadeghian, Amir, Reid, Ian, Savarese, Silvio, 2019. Generalized intersection over union: A metric and a loss for bounding box regression. In: *Paper Presented at the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Rodriguez, A.L., Mikolajczyk, K., 2019. Domain adaptation for object detection via style consistency. *arXiv preprint arXiv:1911.10033*.
- Rombach, R., Blattmann, A., Lorenz, D., Esser, P., Ommer, B., 2022. High-resolution image synthesis with latent diffusion models. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10684–10695.
- RoyChowdhury, Aruni, Chakrabarty, Prithvijit, Singh, Ashish, Jin, SouYoung, Jiang, Huaizu, Cao, Liangliang, Learned-Miller, Erik, 2019. Automatic adaptation of object detectors to new domains using self-training. In: *Paper Presented at the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- RV, Nachiketh, Krishnan, Alok, Krishnan, K., Vinay, Haritha Z.A., Alias, Sasinas, 2021. Southern Pea/Weed field image dataset for semantic segmentation and crop/weed classification using an encoder-decoder network. In: *Paper Presented at the Proceedings of the International Conference on Systems, Energy & Environment (ICSEE)*.
- Sa, Inkyu, Chen, Zetao, Popović, Marija, Khanna, Raghav, Liebsch, Frank, Nieto, Juan, Siegwart, Roland, 2017. Weednet: dense semantic weed classification using multispectral images and mav for smart farming. *IEEE Robot. Automat. Lett.* 3 (1), 588–595.
- Sa, Inkyu, Popović, Marija, Khanna, Raghav, Chen, Zetao, Lottes, Philipp, Liebsch, Frank, Nieto, Juan, Stachniss, Cyrill, Walter, Achim, Siegwart, Roland, 2018. WeedMap: a large-scale semantic weed mapping framework using aerial multispectral imaging and deep neural network for precision farming. *Remote Sens.* 10 (9), 1423.
- Salazar-Gomez, Adrian, Darbyshire, Madeleine, Gao, Junfeng, Sklar, Elizabeth I., Parsons, Simon, 2022. Beyond mAP: Towards practical object detection for weed spraying in precision agriculture. In: *Paper Presented at the 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.
- Shao, Shuai, Li, Zeming, Zhang, Tianyuan, Peng, Chao, Yu, Gang, Zhang, Xiangyu, Li, Jing, Sun, Jian, 2019. Objects365: A large-scale, high-quality dataset for object detection. In: *Paper Presented at the Proceedings of the IEEE/CVF International Conference on Computer Vision*.
- Steininger, Daniel, Trondl, Andreas, Croonen, Gerardus, Simon, Julia, Widhalm, Verena, 2023. The CropAndWeed dataset: a multi-modal learning approach for efficient crop and weed manipulation. In: *Paper Presented at the Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*.
- Sudars, Kaspars, Jasko, Janis, Namatevs, Ivars, Ozola, Liva, Badaukis, Niks, 2020. Dataset of annotated food crops and weed images for robotic computer vision control. *Data Brief* 31, 105833.
- Teimouri, Nima, Dyrmann, Mads, Nielsen, Per Rydahl, Mathiasen, Solvejg Kopp, Somerville, Gayle J., Jørgensen, Rasmus Nyholm, 2018. Weed growth stage estimator using deep convolutional neural networks. *Sensors* 18 (5), 1580.
- Teimouri, Nima, Jørgensen, Rasmus Nyholm, Green, Ole, 2022. Novel assessment of region-based CNNs for detecting monocot/dicot weeds in dense field environments. *Agronomy* 12 (5), 1167.
- Tobin, Josh, Fong, Rachel, Ray, Alex, Schneider, Jonas, Zaremba, Wojciech, Abbeel, Pieter, 2017. Domain randomization for transferring deep neural networks from simulation to the real world. In: *Paper Presented at the 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.
- Wang, Pei, Tang, Yin, Luo, Fan, Wang, Lihong, Li, Chengsong, Niu, Qi, Li, Hui, 2022. Weed25: a deep learning dataset for weed identification. *Front. Plant Sci.* 13, 1053329.
- Wang, Su, Saharia, Chitwan, Montgomery, Ceslee, Pont-Tuset, Jordi, Noy, Shai, Pellegrini, Stefano, Onoe, Yasumasa, Laszlo, Sarah, Fleet, David J., Soricut, Radu, 2023. Image editor and editbench: Advancing and evaluating text-guided image inpainting. In: *Paper presented at the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Weyler, Jan, Läbe, Thomas, Magistri, Federico, Behley, Jens, Stachniss, Cyrill, 2023. Towards domain generalization in crop and weed segmentation for precision farming robots. *IEEE Robot. Automat. Lett.* 8 (6), 3310–3317.
- Xu, Ke, Jiang, Zhijian, Liu, Qihang, Xie, Qi, Zhu, Yan, Cao, Weixing, Ni, Jun, 2022. Multi-modal and multi-view image dataset for weeds detection in wheat field. *Front. Plant Sci.* 13, 936748.
- Yang, X., Song, Z., King, I., Xu, Z., 2022. A survey on deep semi-supervised learning. *IEEE Trans. Knowl. Data Eng.* 35 (9), 8934–8954.
- Yu, Jialin, Schumann, Arnold W., Cao, Zhe, Sharpe, Shaun M., Boyd, Nathan S., 2019. Weed detection in perennial ryegrass with deep learning convolutional neural network. *Front. Plant Sci.* 10, 1422.
- Zhang, D., Li, J., Xiong, L., Lin, L., Ye, M., Yang, S., 2019. Cycle-consistent domain adaptive faster RCNN. *IEEE Access* 7, 123903–123911.
- Zhang, Haoyang, Wang, Ying, Dayoub, Feras, Sunderhauf, Niko, 2021. Varifocalnet: An iou-aware dense object detector. In: *Paper presented at the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Zhang, Hao, Li, Feng, Liu, Shilong, Zhang, Lei, Su, Hang, Zhu, Jun, Ni, Lionel M., Shum, Heung-Yeung, 2022. Dino: Detr with improved denoising anchor boxes for end-to-end object detection. *arXiv*. <https://arxiv.org/abs/2203.03605>. preprint arXiv:2203.03605.
- Zhang, Lvmin, Rao, Anyi, Agrawala, Maneesh, 2023. Adding conditional control to text-to-image diffusion models. In: *Paper presented at the Proceedings of the IEEE/CVF International Conference on Computer Vision*.
- Zhao, Ganlong, Li, Guanbin, Xu, Ruijia, Lin, Liang, 2020. Collaborative training between region proposal localization and classification for domain adaptive object detection. In: *Paper Presented at the Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020. Proceedings, Part XVIII*, p. 16.
- Zheng, Yangtao, Huang, Di, Liu, Songtao, Wang, Yunhong, 2020. Cross-domain object detection through coarse-to-fine feature adaptation. In: *Paper presented at the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Zheng, Zhaoxui, Wang, Ping, Ren, Dongwei, Liu, Wei, Ye, Rongguang, Hu, Qinghua, Zuo, Wangmeng, 2021. Enhancing geometric factors in model learning and inference for object detection and instance segmentation. *IEEE Trans. Cybernet.* 52 (8), 8574–8586.
- Tian, Zhi, Shen, Chunhua, Chen, Hao, He, Tong, 2019. FCOS: Fully convolutional one-stage object detection. *Paper presented at the Proceedings of the IEEE/CVF International Conference on Computer Vision*.
- Heap, I., 2024. The International Herbicide-Resistant Weed Database. Available at: www.weedscience.org.